VEPOLINK
MONITORING PURITY

PROCESS INSTRUMENTATION × FLOW MEASUREMENT

CONTINUOUS FLOW MEASUREMENT FOR CONDUCTIVE LIQUIDS

EMF Flowmeter *Inline Electromagnetic*

A full-bore electromagnetic flowmeter built on Faraday's law, for water, wastewater, slurries and process liquids — hermetically sealed, bidirectional, and sized from DN10 to DN3000 and beyond.



10–3000 mm
NOMINAL BORE, DN10 TO DN3000+

0.2–12 m/s
BIDIRECTIONAL VELOCITY RANGE

±0.5 %
ACCURACY OF MEASURED VALUE

FARADAY ELECTROMAGNETIC INDUCTION | **INTEGRAL / REMOTE** TRANSMITTER |
VEPOLINK FIELD-PROVEN BUILD

>20 µS/cm
MINIMUM LIQUID CONDUCTIVITY

±0.5%
ACCURACY OF MEASURED VALUE

10 – 3000mm
NOMINAL BORE RANGE

IP68
SUBMERSIBLE SENSOR OPTION

01 OVERVIEW

Flow, measured *without moving parts.*

A full-bore electromagnetic flowmeter for any electrically conductive liquid above 20 µS/cm — water, wastewater, sludge, slurries, pastes, acids, alkalis, juices and fruit pulp — with or without suspended solids. There is no obstruction in the flow path, so there is nothing to wear, clog or add pressure drop.

The coil assembly is hermetically sealed in a welded construction for long-term reliability, and pulsed DC excitation keeps the zero point stable and rejects electrochemical noise. The transmitter is available integral to the sensor or mounted remotely, with field-interchangeable electronics.

REPORTED CHANNELS

- INSTANTANEOUS FLOW RATE
- FORWARD TOTALISER
- REVERSE TOTALISER
- NET TOTALISER
- 4-20 MA OUTPUT
- PULSE / FREQUENCY OUTPUT
- EMPTY PIPE DETECTION
- FLOW DIRECTION
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

Faraday's Law Principle

Velocity is derived from the EMF induced as a conductive liquid moves through a magnetic field — a direct, obstruction-free measurement.

Hermetically Sealed Coils

A welded coil assembly keeps moisture and process vapour out, holding calibration stable over a long service life.

Integral or Remote

Mount the transmitter directly on the sensor, or remotely on a field mount kit up to 10 m away, with field-interchangeable electronics.

Pulsed DC Excitation

Alternating excitation polarity gives absolute zero stability and eliminates electrochemical and process noise from the reading.

02 MEASUREMENT PRINCIPLE

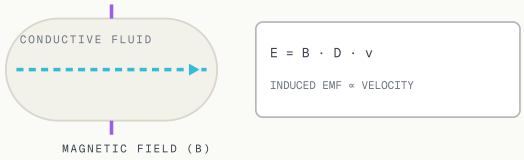
How electromagnetic *induction measures flow.*

A magnetic field is applied across the flow tube. As the conductive liquid moves through it, it generates a small voltage picked up by a pair of electrodes — proportional to velocity, with no moving parts in the flow path.

FARADAY

Electromagnetic Induction

MAGNETIC COILS · CONDUCTIVE FLUID · INDUCED EMF



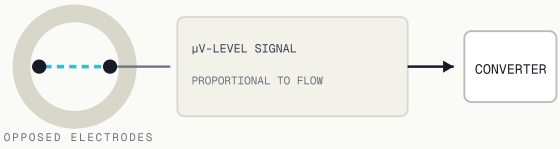
A magnetic field (B) is generated across the tube bore. A conductive liquid moving through it at velocity (v) induces a voltage (E) directly proportional to flow velocity.

`E = B · D · v`

PICKUP

Electrode Signal Pickup

OPPOSED ELECTRODES · MV-LEVEL SIGNAL



A pair of electrodes mounted opposite each other, flush with the tube's non-conductive liner, picks up the induced voltage and passes it to the converter electronics.

`Signal ∝ flow velocity`

STABILITY

Pulsed DC Excitation

ALTERNATING POLARITY · ZERO REJECTION



The excitation current alternates polarity at a controlled rate, letting the converter separate the true flow signal from zero drift, electrochemical noise and stray process potentials.

`Absolute zero stability`

OUTPUT

Conversion & Totalising

4-20 MA · PULSE · LOCAL DISPLAY



The converter linearises the signal into rate and volume, shown on a local 2-line LCD and passed on as an analog, pulse or digital output to SCADA, PLC or batching systems.

`Rate · forward · reverse · net totals`

03 SENSOR CONSTRUCTION

What's inside the *flow tube.*

Everything between the process liquid and the converter output is built to resist wear, hold calibration, and keep the measurement unaffected by what is flowing.

STEP 01

Non-conductive liner

A rubber (or PTFE for high-temperature duty) liner isolates the measuring tube from the process liquid and the magnetic field from the metal body.

STEP 02

Field coils energised

The hermetically sealed, welded coil assembly generates a controlled, pulsed DC magnetic field across the full bore of the tube.

STEP 03

Electrodes sense EMF

SS316L electrodes, flush with the liner, pick up the µV-level induced voltage as the liquid passes through the field.

STEP 04

Convert and report

The integral or remote converter linearises, displays, totalises and transmits the signal to the plant system.

NO OBSTRUCTION, NO WEAR

Because the field coils and electrodes sit outside the flow path, there is nothing protruding into the liquid to wear, clog, or add pressure drop — a meaningful advantage on slurries, sludges and liquids carrying solids.

MATERIAL SELECTION

Measuring tube in SS304 as standard, carbon steel sensor housing, and SS316L electrodes as standard — with liner and electrode material selectable to suit abrasive, corrosive or hygienic duty.

WHY THIS MEASUREMENT MATTERS

Accurate, repeatable flow measurement underpins billing, mass balance, dosing and regulatory reporting across water, wastewater and process plants. An electromagnetic flowmeter gives that measurement on conductive liquids — including dirty, viscous or solids-laden streams — without a moving part to wear out or recalibrate against.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

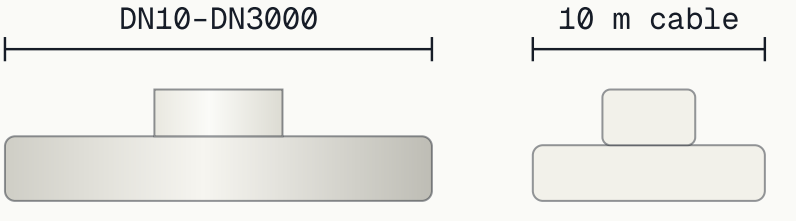
Parameter	Volumetric flow, conductive liquids
Working Principle	Faraday's law, electromagnetic induction
Nominal Diameter	10 - 3000 mm, and above
Measuring Range	0.2 - 12 m/s, bidirectional
Minimum Conductivity	> 20 µS/cm
Accuracy	±0.5% of measured value (±0.2% on request)
Repeatability	±0.2% of span
Linearity	High, by design of magnetic field profile
Working Pressure	10 / 16 / 25 / 40 kg/cm²
Working Temperature	Integral PTFE 120 °C · Remote PTFE 180 °C · others 70 °C

SYSTEM · BUILD · POWER

Display	2-line LCD, local units (m³, litre, gallon, ft³)
Transmitter	Integral or remote, field-interchangeable electronics
Excitation	Pulsed DC · absolute zero stability
Measuring Tube	SS304, standard
Sensor Housing	Carbon steel, standard
Electrode Material	SS316L, standard
Sensor Lining	Rubber, standard (PTFE option)
Output	4-20 mA, standard; pulse / frequency option
Power Supply	12-60 VDC or 80-300 VAC/DC · solar powered option
Protection Class	Sensor IP65 (IP67 / IP68 option) · transmitter IP67
End Connection	Flange / wafer / tri-clamp / SMS · ANSI 150 standard
Remote Cable Length	10 m, standard

INSTRUMENT DIMENSIONS

INDICATIVE SENSOR & REMOTE CONVERTER · NOT TO SCALE



05 INSTALLATION & UPKEEP

Flanged in, *calibrated, left running.*

> The sensor is installed inline, full bore, with the flow tube always kept full of liquid. Each unit is factory calibrated at a gravimetric **Water Flow Calibration Facility**, under controlled and stable flow conditions, before dispatch.

01 · ORIENTATION

Always full bore

Install with the tube always full — vertical upflow or a low point in a horizontal run — to avoid false readings from a part-full pipe.

02 · STRAIGHT RUNS

Upstream / downstream

Keep adequate straight pipe upstream and downstream of the sensor, clear of valves, bends and pumps, to settle the velocity profile.

03 · GROUNDING

Bonded to the process

Ground rings or straps bond the liner's process-wetted liquid to the pipe potential, keeping the electrode signal clean.

04 · TRANSMITTER MOUNT

Integral or field remote

Mount the converter on the sensor for compact installs, or remotely up to the standard 10 m cable length where vibration or access demands it.

05 · CALIBRATION

Factory traceable

Each meter ships with a calibration record from the gravimetric facility; periodic verification follows the site's instrument audit schedule.

06 · DIAGNOSTICS

Reported with the data

Empty-pipe detection, flow direction and fault status travel alongside the rate and totaliser outputs.

SIZING A METER

Nominal diameter is selected to keep expected flow within the 0.2–12 m/s velocity window at the chosen bore, with liner, electrode and end-connection selected for the liquid's chemistry, temperature and the plant's piping standard. Our team sizes the meter from your flow range and process conditions.

COMMISSIONING

Commissioning covers grounding verification, an empty-pipe and zero check with the line full and static, and confirmation of output scaling against the plant's control or billing system — after which the meter runs unattended, with no consumables and no moving parts to service.

06 APPLICATIONS

Where it *works.*

Any electrically conductive liquid, clean or carrying solids, is a candidate — from municipal water billing to the most demanding process and effluent duties.

Chemical & Petrochemical	Food & Drug Industries	Paper & Pulp
Dredging Industries	Fertiliser Industries	Pharmaceutical
Sugar & Beverage	Water & Wastewater	Mining Industries

CLEAN OR DIRTY, SAME MEASUREMENT

Because the measurement has no moving part and no restriction in the bore, the same operating principle covers potable water billing and abrasive slurry or sludge duty — the difference is in liner and electrode material, not in the method.

BILLING & MASS BALANCE

Forward, reverse and net totalisers make the meter suitable for bidirectional billing points, while the 4–20 mA and pulse outputs feed plant mass-balance and process-control loops directly.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Faraday's law of electromagnetic induction. A magnetic field is applied across the flow tube; the conductive liquid moving through it induces a voltage proportional to its velocity.

Q What liquids can it measure?

Any electrically conductive liquid above roughly 20 µS/cm — water, wastewater, sludge, slurries, pastes, acids, alkalis, juices and fruit pulp — with or without suspended solids.

Q What sizes are available?

Flow tube sizes from 10 mm up to 3000 mm and above, sized to keep the expected flow within the 0.2–12 m/s velocity window.

Q Can it measure flow in both directions?

Yes, it is bidirectional, reporting forward, reverse and net totals from the same sensor.

Q Does the pipe need to run full?

Yes, the measuring tube must stay full of liquid for an accurate reading. Install at a low point or in a vertical upflow run to assure this.

Q Integral or remote transmitter — which do I need?

Integral mount suits most installs; remote mount (standard 10 m cable) suits sensors in pits, under insulation, or subject to vibration or heat that the electronics should avoid.

Q How is zero stability maintained?

Pulsed DC excitation alternates polarity, letting the converter reject zero drift and electrochemical noise for absolute zero stability.

Q What outputs are available?

A local 2-line LCD display, a standard 4–20 mA analog output, and pulse/frequency output for integration with SCADA, PLC or batching systems.

Q Is it calibrated before dispatch?

Yes, each meter is factory calibrated at a gravimetric water flow calibration facility under controlled, stable flow conditions.

Q What maintenance does it need?

Very little — there are no moving parts or consumables in the flow path. Periodic grounding and zero checks per the site's audit schedule are all that is typically required.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

FLOW, MEASURED WITH CONFIDENCE

Know your flow. *In real time.*

Tell us the liquid, the bore and the pressure rating, and we'll configure the sensor, liner, electrode material and transmitter mount around your process. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that sizes, specifies and commissions electromagnetic flowmeters. No call centre.
+91 99538 86178 →

02 **Request a sizing & configuration**
RECOMMENDED
Send your bore, liquid and pressure rating — we'll spec liner, electrodes, transmitter mount and output.
sales@vepolink.com →

03 **Book a demonstration**
See a running meter and the converter output with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM